

Unit 8, Lesson 9: Nets of Rectangular Prisms and Pyramids

Benchmark

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

Learning Targets

- ☐ I can draw the net of a rectangular prism or rectangular pyramid.
- ☐ I can relate a net to its three-dimensional shape.
- ☐ I can relate a rectangular prism or rectangular pyramid to its net.

8.9.1 Warm-Up: Bell Work

1. A rectangular ball pit has a volume of 28 cubic meters (m^3). Find the width of the ball pit if the length is $5\frac{5}{6}$ m and the depth is $\frac{3}{4}$ m.

8.9.2 Refresher: Area of Composite Figures

1. Work with your partner to find the area of each composite figure. All measurements are in meters (m).

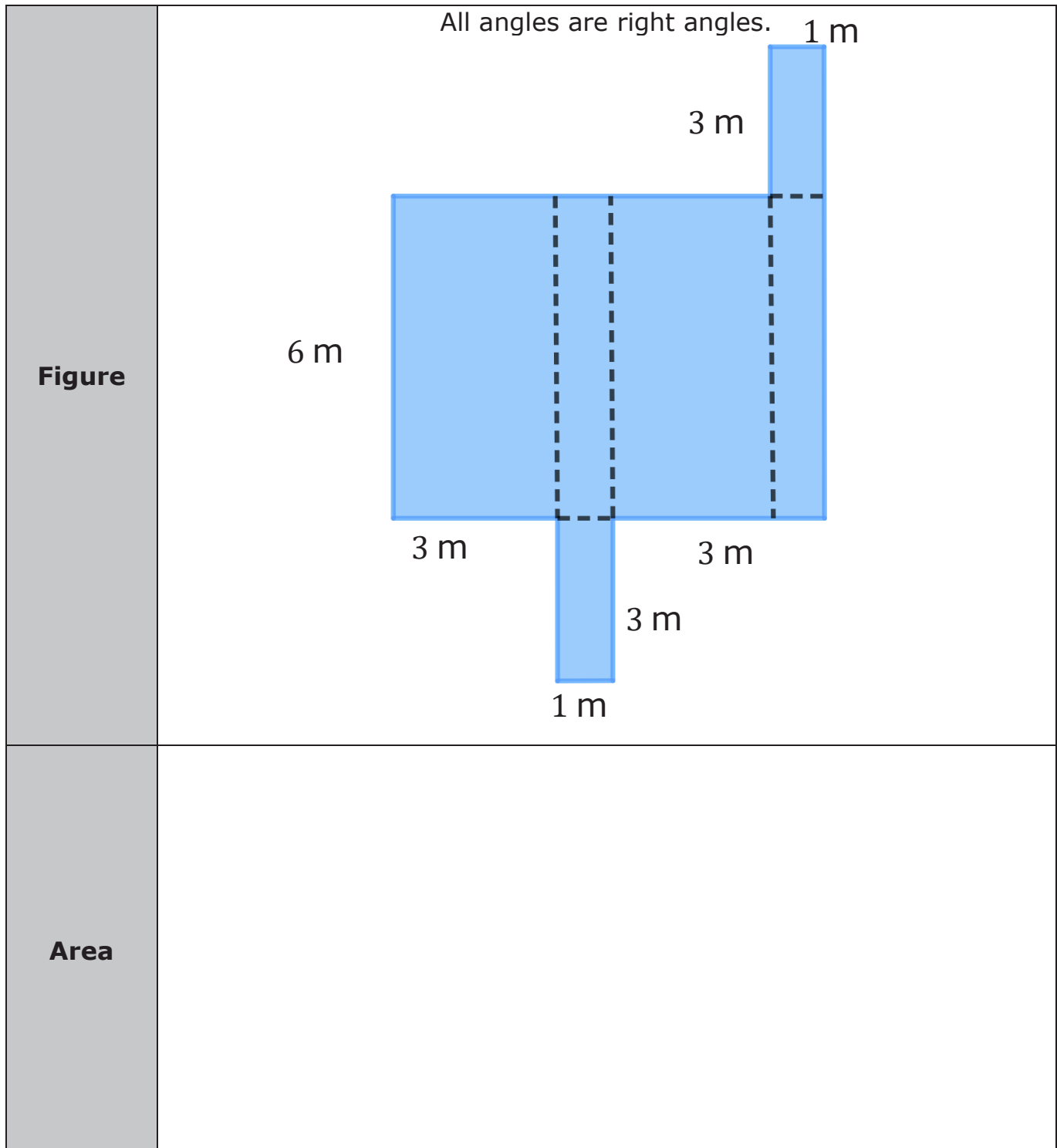
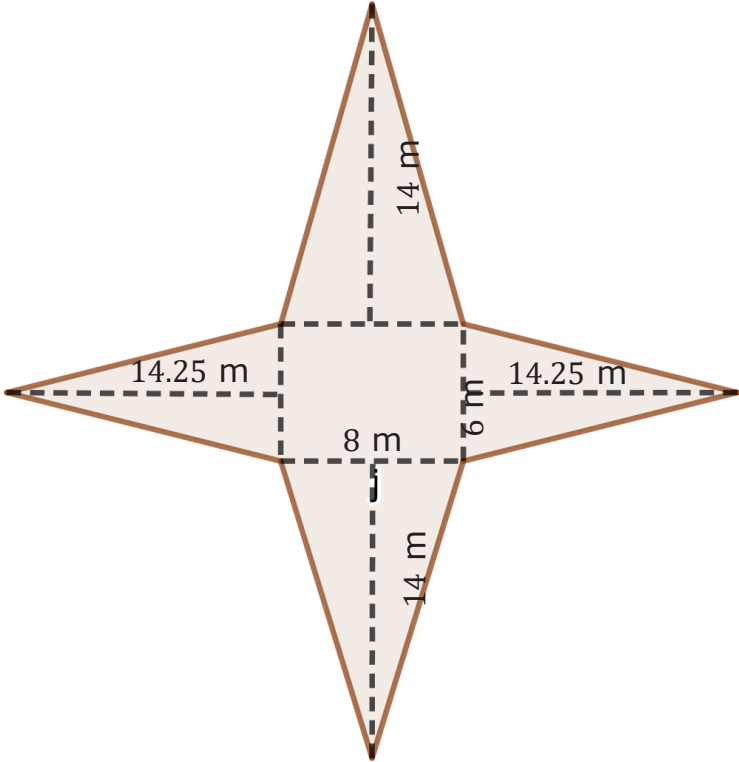


Figure	All dashed lines intersect at right angles.  The figure is a four-pointed star. It consists of a central square with a side length of 8 m. Four triangles are attached to the sides of the square. Each triangle has a base of 6 m and a height of 14 m. The distance from the center of the star to the tip of each point is 14.25 m. All dashed lines intersect at right angles.
Area	

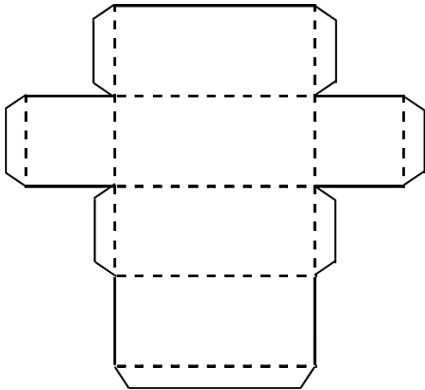
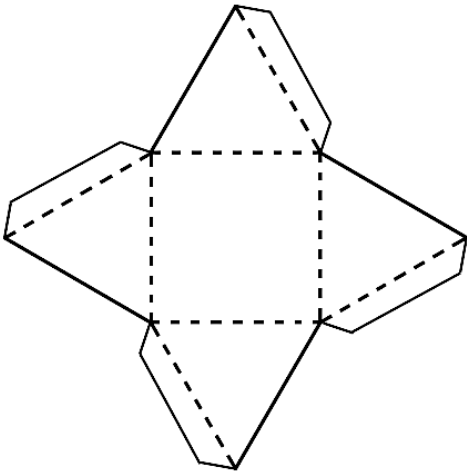
8.9.3 Exploration: Creating Three-Dimensional Figures

Both composite figures in the previous component are nets. A net is formed when the faces of a three-dimensional figure are positioned by arranging the faces to create a two-dimensional composite figure.



A **net** is a two-dimensional diagram that can be folded or made into a three-dimensional figure.

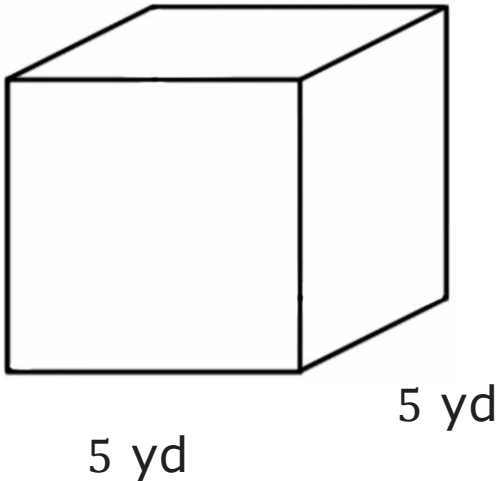
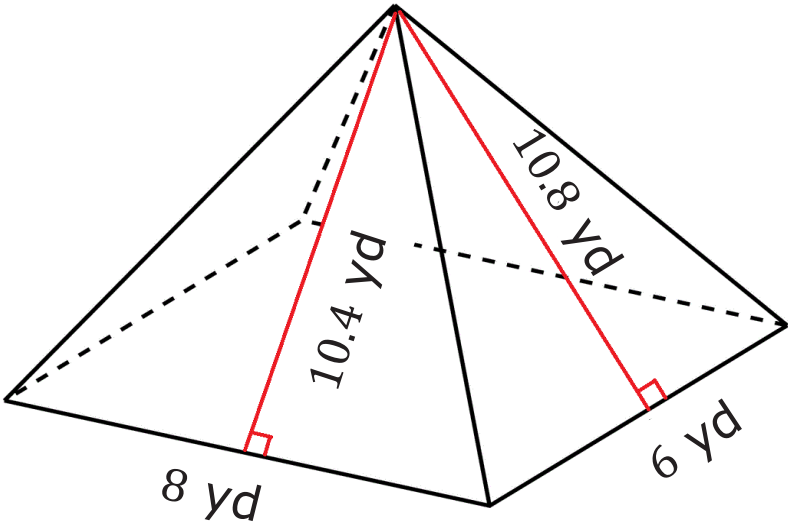
- 1. Your teacher will provide you with copies of some nets. Cut the nets along the solid line. Fold the nets along the dotted line and use the tabs to glue the figure together.
- 2. Complete the table to describe the three-dimensional figure formed.

Net	Three-Dimensional Figure Formed
	
	

3. Look back at the two composite figures from the refresher. Predict what three-dimensional figure is made from cutting out and folding up the sides of those figures.
4. With your partner, discuss how the area of a two-dimensional composite figure is related to the area of the faces of the three-dimensional figure. Summarize your discussion.

8.9.4 Collaboration: Determining Nets

1. With your partner, sketch the net or sketch the individual faces of the three-dimensional figure. Label each side of the net with the appropriate measurement in yards (yd).

Three-Dimensional Figure	Net
	
	

8.9.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on nets?

